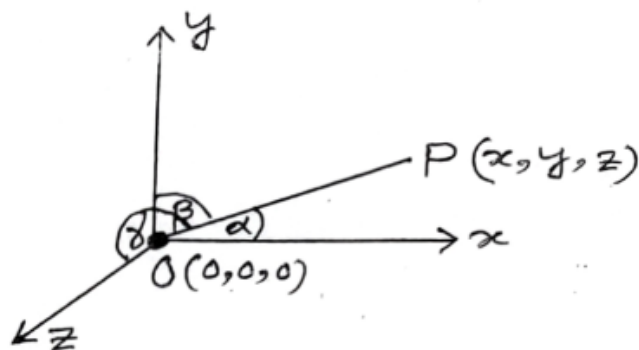


Rectangular Coordinates

Direction cosines of a line



If a given line OP makes angles α, β, γ with the positive direction of x, y, z -axes, respectively, then $\cos \alpha, \cos \beta, \cos \gamma$ are the *direction cosines* (dcs) of the line OP and are generally denoted by the letters l, m, n , respectively.

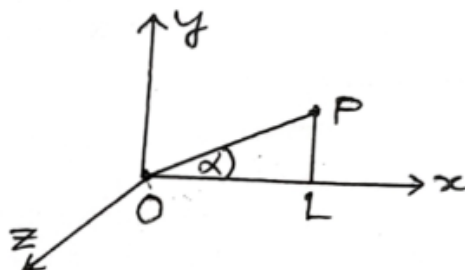
Again, the dcs of PO will be $\cos(\pi + \alpha), \cos(\pi + \beta), \cos(\pi + \gamma)$ because the line through O , i.e., PO will now makes angles $\pi + \alpha, \pi + \beta, \pi + \gamma$ with the coordinate axes.

If a line does not pass through the origin O , then its direction angles are the angles between the axes and a line drawn through O parallel to the given line (keeping the direction unchanged).

Direction cosines of the axes of references

Since the x -axis makes angles $0^\circ, 90^\circ, 90^\circ$ with the coordinate axes, its dcs are $\cos 0, \cos 90, \cos 90$, i.e., $(1, 0, 0)$. Similarly, the dcs of y and z -axes are respectively $(0, 1, 0)$ and $(0, 0, 1)$.

If O be the origin and $P(x, y, z)$ be any point, then $x = lr, y = mr, z = nr$, where l, m, n are the dcs of OP and r is the length of OP .



Through P draw PL perpendicular to x -axis so that $OL = x$. Now from $\triangle OPL$, we have

$$\frac{OL}{OP} = \cos \angle LOP = \cos \alpha \Rightarrow \frac{x}{r} = \cos \alpha \Rightarrow x = lr.$$

Similarly, we can show that $y = mr$ and $z = nr$ by drawing perpendiculars to y and z -axes, respectively. So the coordinates of P are (lr, mr, nr) .

Note: We have $x = lr, y = mr$ and $z = nr$ from which we get

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = r.$$

Relation between direction cosines

If l, m, n be the direction cosines of any line, then

$$l^2 + m^2 + n^2 = 1.$$

Let OP be drawn through the origin parallel to the given line so that l, m, n are the dcs of angles which the line makes with OX, OY, OZ , respectively.

Let $P(x, y, z)$ be the coordinates of P and $OP = r$. Then we have $x = lr, y = mr$ and $z = nr$.

$$\begin{aligned}\text{So, } x^2 + y^2 + z^2 &= (l^2 + m^2 + n^2)r^2 \\ \Rightarrow r^2 &= (l^2 + m^2 + n^2)r^2 \\ \Rightarrow l^2 + m^2 + n^2 &= 1, \text{ i.e., } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.\end{aligned}$$

Direction ratios

Any three numbers a, b, c which are proportional to the dcs l, m, n , respectively, are called the *direction ratios* (drs) of the given line.

By definition, we can write

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}$$

$$\text{Therefore, } l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}.$$

Hence if (a, b, c) be three numbers proportional to the direction cosines of OP , then the actual

$$\text{dcs of } OP \text{ are } \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}},$$

and that of PO are

$$\frac{-a}{\sqrt{a^2 + b^2 + c^2}}, \frac{-b}{\sqrt{a^2 + b^2 + c^2}}, \frac{-c}{\sqrt{a^2 + b^2 + c^2}}.$$

Projection of a point on a straight line

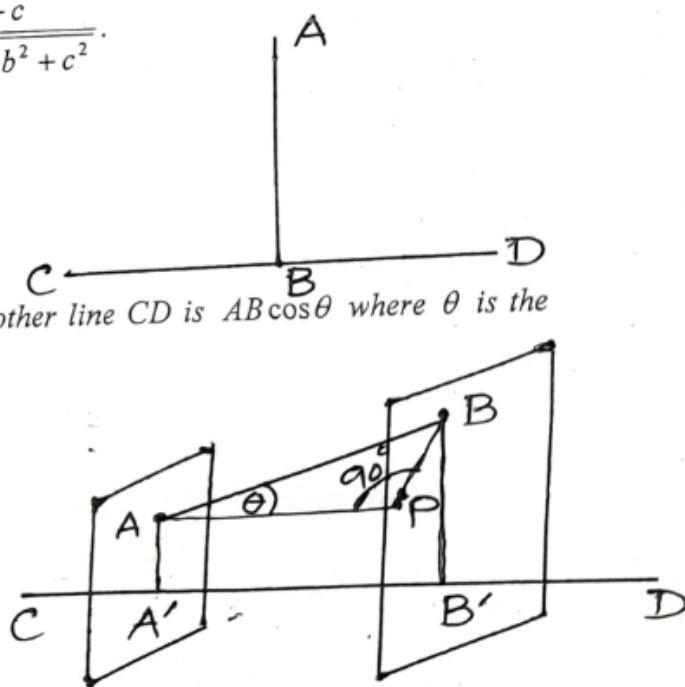
The projection of a point A on a given line CD is the point B which is the foot of the perpendicular from A on CD .

The orthogonal projection of a given line AB on another line CD is $AB \cos \theta$ where θ is the angle between AB and CD .

Through A draw AP parallel to $A'B'$ and meeting the plane through B at P . So $AP = A'B'$. Now if θ be the angle between AB and CD , then AP being parallel to CD , we have $\angle PAB = \theta$.

Also, since PB lies in the plane perpendicular to AP , so that $\angle APB = 90^\circ$ and $AP = AB \cos \theta$.

But $A'B'PA$ is a rectangle so that $AP = A'B'$. Thus $A'B' = AP = AB \cos \theta$.



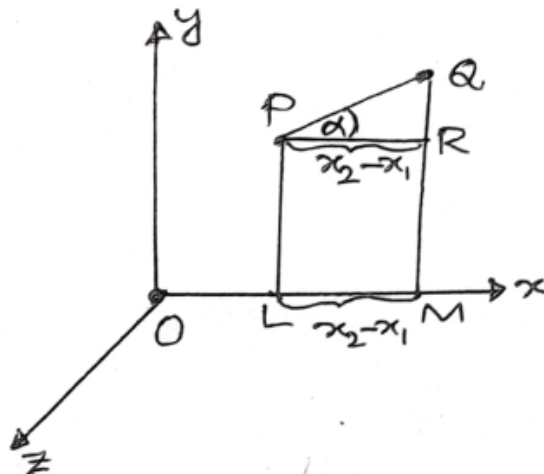
Direction cosines of a line joining points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Let L, M be the feet of the perpendicular drawn from P, Q to the x -axis so that $OL = x_1$ and $OM = x_2$.

So projection of PQ on x -axis is $LM = OM - OL = x_2 - x_1$.

Also, if l, m, n be the dcs of PQ , then the projection of PQ on x -axis is given by $lPQ = x_2 - x_1$.

Similarly, the projection of PQ on y and z -axes are respectively $mPQ = y_2 - y_1$ and $nPQ = z_2 - z_1$.



Therefore, $\frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n} = PQ$

Thus, the dcs of the line joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$.

Projection of a broken line consisting of continuous segments on a given line.

Let $P_1, P_2, \dots, P_n$ be n number of points in space and $Q_1, Q_2, \dots, Q_n$ be their projections on the line AB , these points being on the same straight line, we have

$$Q_1 Q_n = \text{projection of } P_1 P_n = Q_1 Q_2 + Q_2 Q_3 + \dots + Q_{n-1} Q_n,$$

where $Q_1 Q_2 = \text{projection of } P_1 P_2$,

$Q_2 Q_3 = \text{projection of } P_2 P_3$, and so on.

Hence the sum of the projections of $P_1 P_2, P_2 P_3, \dots, P_{n-1} P_n$ on AB

$$= \text{projection of } P_1 P_n \text{ on } AB$$

$$= Q_1 Q_n.$$

Projection of a line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line with dcs l, m, n is

$$(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n.$$

Through P and Q draw planes parallel to the coordinate planes to form a rectangular parallelepiped with one diagonal PQ .

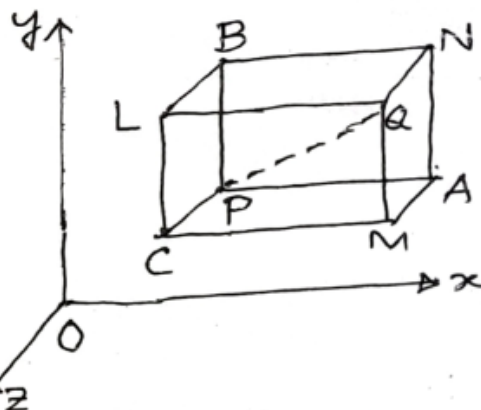
$$\text{So } PA = x_2 - x_1, AN = y_2 - y_1, NQ = z_2 - z_1.$$

If α, β, γ be the angles which the line makes with the coordinate axes, then $l = \cos \alpha, m = \cos \beta$, and $n = \cos \gamma$.

Hence the projection PA on the line is $PA \cos \alpha = (x_2 - x_1)l$.

Similarly, the projection of AN on the line is $AN \cos \beta = (y_2 - y_1)m$ and the projection of NQ on the line is $NQ \cos \gamma = (z_2 - z_1)n$.

But the sum of the projections of PA, AN and NQ on the line is equal to the projection of PQ on that line and hence the required projection is $(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n$.



Angle between two lines

If θ be the angle between the lines with dcs (l_1, m_1, n_1) and (l_2, m_2, n_2) , then

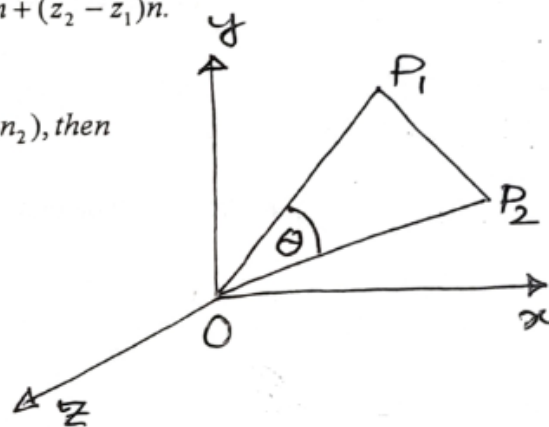
$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2.$$

Let OP and OQ be two lines through O whose dcs are (l_1, m_1, n_1) and (l_2, m_2, n_2) , and the angle between OP and OQ is θ .

Let P be the point (x_1, y_1, z_1) . Draw PL perpendicular to the xy -plane and LA , perpendicular to the x -axis.

Then the projection of OP on $OQ = \text{sum of the projection of } OA, AL, \text{ and } LP \text{ on } OQ$

$$OP \cos \theta = x_1 l_2 + y_1 m_2 + z_1 n_2$$



But x_1 = projection of OP on x -axis = $OP \cdot l_1$

y_1 = projection of OP on y -axis = $OP \cdot m_1$

z_1 = projection of OP on z -axis = $OP \cdot n_1$

Thus, $OP \cos \theta = (OP)l_1l_2 + (OP)m_1m_2 + (OP)n_1n_2$

Hence $\cos \theta = l_1l_2 + m_1m_2 + n_1n_2$.

Corollary: We have $\cos \theta = l_1l_2 + m_1m_2 + n_1n_2$.

$$\begin{aligned}\text{So that } \sin^2 \theta &= 1 - \cos^2 \theta = 1 - (l_1l_2 + m_1m_2 + n_1n_2)^2 \\ &= (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2 \\ &= (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 \\ &= \Sigma (m_1n_2 - m_2n_1)^2.\end{aligned}$$

Therefore, $\sin \theta = \pm \sqrt{\Sigma (m_1n_2 - m_2n_1)^2}$.

Corresponding formula when direction ratios of the lines are given

If a_1, b_1, c_1 and (a_2, b_2, c_2) be proportional to the dcs of the line OP_1 and OP_2 , then their actual

dcs are $l_1 = \pm \frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_1}{\sqrt{\Sigma a_1^2}}$, $m_1 = \pm \frac{b_1}{\sqrt{\Sigma a_1^2}}$, and $n_1 = \pm \frac{c_1}{\sqrt{\Sigma a_1^2}}$;

$$l_2 = \pm \frac{a_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}} = \pm \frac{a_2}{\sqrt{\Sigma a_2^2}}, \quad m_2 = \pm \frac{b_2}{\sqrt{\Sigma a_2^2}}, \quad \text{and } n_2 = \pm \frac{c_2}{\sqrt{\Sigma a_2^2}}.$$

If θ be the angle between them, then $\cos \theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\Sigma a_1^2} \times \sqrt{\Sigma a_2^2}}$

$$\text{and } \sin \theta = \pm \frac{\sqrt{(b_1c_2 - b_2c_1)^2 + (a_1b_2 - a_2b_1)^2 + (c_1a_2 - c_2a_1)^2}}{\sqrt{\Sigma a_1^2} \times \sqrt{\Sigma a_2^2}} = \pm \frac{\sqrt{\Sigma (b_1c_2 - b_2c_1)^2}}{\sqrt{\Sigma a_1^2} \times \sqrt{\Sigma a_2^2}}.$$

Condition for perpendicularity of two lines

If the lines OP_1 and OP_2 are perpendicular to each other, then $\cos \theta = 0$ and so

$$l_1l_2 + m_1m_2 + n_1n_2 = 0.$$

Using direction ratios, the equation can be put as $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

Condition for parallelism of two lines

If the lines OP_1 and OP_2 be parallel to each other, then $\sin \theta = 0$ and so the condition is

$$\begin{aligned}(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 &= 0 \\ \Rightarrow l_1m_2 - l_2m_1 = 0, m_1n_2 - m_2n_1 = 0 \text{ and } n_1l_2 - n_2l_1 &= 0 \\ \Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{l_2^2 + m_2^2 + n_2^2}} &= 1\end{aligned}$$

Hence $l_1 = l_2, m_1 = m_2, n_1 = n_2$.

Using direction ratios, we get

$$\begin{aligned} & (b_1c_2 - b_2c_1)^2 + (a_1b_2 - a_2b_1)^2 + (c_1a_2 - c_2a_1)^2 = 0 \\ \Rightarrow & b_1c_2 - b_2c_1 = 0, a_1b_2 - a_2b_1 = 0 \text{ and } c_1a_2 - c_2a_1 = 0 \\ \Rightarrow & \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{a_1}{a_2} \end{aligned}$$

Worked out problems

1. Show that the straight lines whose direction cosines are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$ are at right angles.

Solution: Given that $2l + 2m - n = 0 \Rightarrow n = 2(l + m)$ (i)

and $mn + nl + lm = 0$

$$\Rightarrow (m + n)2(l + m) + lm = 0$$

$$\Rightarrow 2l^2 + 5lm + 2m^2 = 0$$

$$\Rightarrow (2l + m)(l + 2m) = 0 \text{ either } 2l + m + 0n = 0 \quad \text{(ii)}$$

$$\text{or } l + 2m + 0n = 0 \quad \text{(iii)}$$

Solving (i) and (ii), we get

$$\frac{l}{1} = \frac{m}{-2} = \frac{n}{-2}$$

So the direction cosines of the first line are proportional to 1, -2, -2.

Similarly, solving (i) and (iii), we get

$$\frac{l}{2} = \frac{m}{-1} = \frac{n}{2}$$

which means that the direction cosines of the second line are proportional to 2, -1, 2.

If θ be the angle between the lines, then

$$\cos \theta = \pm \frac{(2)(1) + (-1)(-2) + (2)(-2)}{\sqrt{4+1+4}\sqrt{1+4+4}} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Thus the given straight lines are at right angles.

2. If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the direction cosines of two mutually perpendicular lines, show that the direction cosines of the line perpendicular to them are

$$m_1n_2 - m_2n_1, n_1l_2 - n_2l_1, l_1m_2 - l_2m_1.$$

Solution: Let (l, m, n) be the direction cosines of the required line. If it is perpendicular to two mutually perpendicular lines with direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) , we have

$$ll_1 + mm_1 + nn_1 = 0 \text{ and } ll_2 + mm_2 + nn_2 = 0. \text{ We can write}$$

$$\frac{l}{m_1n_2 - m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1}.$$

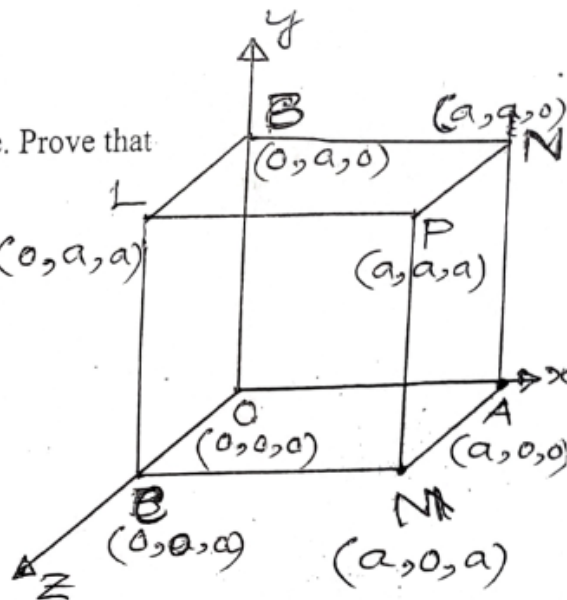
Hence the actual direction cosines of the required line are

Worked out problems

1. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}.$$

Solution: Let a be the length of each side of the cube. Referring to three coterminous edges OA, OB, OC as coordinate axes, the coordinates of the vertices are shown in the adjacent figure.



Now the direction cosines of the diagonal OP

are $\left(\frac{a}{\sqrt{a^2 + a^2 + a^2}}, \frac{a}{\sqrt{a^2 + a^2 + a^2}}, \frac{a}{\sqrt{a^2 + a^2 + a^2}} \right)$, i.e., $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$. Similarly, direction

cosines of AL, BM and CN are $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$, and $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$,

respectively.

Suppose (l, m, n) are the direction cosines of a line which makes angles $\alpha, \beta, \gamma, \delta$ with OP, AL, BM, CN , respectively. Then

$$\cos \alpha = \frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l + m + n),$$

$$\cos \beta = -\frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(-l + m + n),$$

$$\cos \gamma = \frac{1}{\sqrt{3}}l - \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l - m + n),$$

$$\cos \delta = \frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m - \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l + m - n).$$

Hence, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta$

$$= \frac{1}{3}[(l + m + n)^2 + (-l + m + n)^2 + (l - m + n)^2 + (l + m - n)^2]$$

$$= \frac{1}{3}[4(l^2 + m^2 + n^2)] = \frac{4}{3}.$$

Condition for parallelism of two lines

If the lines OP_1 and OP_2 be parallel to each other, then $\sin \theta = 0$ and so the condition is

$$(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 = 0$$

$$\Rightarrow l_1 m_2 - l_2 m_1 = 0, m_1 n_2 - m_2 n_1 = 0 \text{ and } n_1 l_2 - n_2 l_1 = 0$$

$$\Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{l_2^2 + m_2^2 + n_2^2}} = 1$$

Hence $l_1 = l_2, m_1 = m_2, n_1 = n_2$.

Using direction ratios, we get

$$(b_1 c_2 - b_2 c_1)^2 + (a_1 b_2 - a_2 b_1)^2 + (c_1 a_2 - c_2 a_1)^2 = 0$$

$$\Rightarrow b_1 c_2 - b_2 c_1 = 0, a_1 b_2 - a_2 b_1 = 0 \text{ and } c_1 a_2 - c_2 a_1 = 0$$

$$\Rightarrow \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{a_1}{a_2}$$

Worked out problems

1. Find the angle between two diagonals of a cube.

Solution: Let OP and AL be the diagonals of the cube. The direction cosines of OP and AL are

$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$, respectively. Let θ be the angle between the diagonals.

$$\text{Then } \cos \theta = \left(\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right) = -\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{1}{3}$$

$$\text{Hence, } \theta = \cos^{-1}\left(\frac{1}{3}\right).$$

2. If a line makes angles α, β, γ with the axes, show that $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.

Solution: We have $l^2 + m^2 + n^2 = 1$, i.e., $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\Rightarrow 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2.$$

3. If P, Q, R, S are the points $(3, 4, 5), (4, 6, 3), (-1, 2, 4)$ and $(1, 0, 5)$, respectively, find the projection of RS on PQ .

Solution: Let l, m, n be the direction cosines of PQ . Then

$$l = \frac{4-3}{\sqrt{(4-3)^2 + (6-4)^2 + (3-5)^2}} = \frac{1}{3}, \quad m = \frac{6-4}{\sqrt{(4-3)^2 + (6-4)^2 + (3-5)^2}} = \frac{2}{3}$$

$$\text{and } n = \frac{3-5}{\sqrt{(4-3)^2 + (6-4)^2 + (3-5)^2}} = \frac{-2}{3}$$

Therefore, by definition, the projection of RS on PQ is

$$(1+1)\left(\frac{1}{3}\right) + (0-2)\left(\frac{2}{3}\right) + (5-4)\left(\frac{-2}{3}\right) = -4/3.$$

4. If P, Q, R, S are the points $(2, 3, -1), (3, 5, -3), (1, 2, 3)$ and $(3, 5, 7)$, respectively, find the angle between PQ and RS .

Solution: Here, $l_1 = \frac{1}{3}, m_1 = \frac{2}{3}, n_1 = \frac{-2}{3}; l_2 = \frac{2}{\sqrt{29}}, m_2 = \frac{3}{\sqrt{29}}, n_2 = \frac{4}{\sqrt{29}}$

Let θ be the angle between PQ and RS . Therefore, by definition,

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2 = (2/3\sqrt{29}) + (6/3\sqrt{29}) - (8/3\sqrt{29}) = 0 \Rightarrow \theta = \pi/2$$

i.e. PQ and RS are perpendicular lines.

5. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}.$$

Solution: Let a be the length of each side of the cube. Referring to three coterminus edges OA, OB, OC as coordinate axes, the coordinates of the vertices are shown in the adjacent figure.

Now the direction cosines of the diagonal OP are

$$\left(\frac{a}{\sqrt{a^2 + a^2 + a^2}}, \frac{a}{\sqrt{a^2 + a^2 + a^2}}, \frac{a}{\sqrt{a^2 + a^2 + a^2}} \right), \text{ i.e., } \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right).$$

Similarly, direction cosines of AL, BM and CN are $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$, and

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right), \text{ respectively.}$$

Suppose (l, m, n) are the direction cosines of a line which makes angles $\alpha, \beta, \gamma, \delta$ with OP, AL, BM, CN , respectively. Then

$$\cos \alpha = \frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l + m + n),$$

$$\cos \beta = -\frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(-l + m + n),$$

$$\cos \gamma = \frac{1}{\sqrt{3}}l - \frac{1}{\sqrt{3}}m + \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l - m + n),$$

$$\cos \delta = \frac{1}{\sqrt{3}}l + \frac{1}{\sqrt{3}}m - \frac{1}{\sqrt{3}}n = \frac{1}{\sqrt{3}}(l + m - n).$$

Hence, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta$

$$= \frac{1}{3}[(l + m + n)^2 + (-l + m + n)^2 + (l - m + n)^2 + (l + m - n)^2]$$

$$= \frac{1}{3}[4(l^2 + m^2 + n^2)] = \frac{4}{3}.$$

Now that the straight lines whose direction cosines are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$ are at right angles.

Solution: Given that $2l + 2m - n = 0 \Rightarrow n = 2(l + m)$ (i)

and so $mn + nl + lm = 0$

$$\Rightarrow (m + n)2(l + m) + lm = 0$$

$$\Rightarrow 2l^2 + 5lm + 2m^2 = 0$$

$$\Rightarrow (2l + m)(l + 2m) = 0 \text{ either } 2l + m + 0n = 0 \quad \text{(ii)}$$

$$\text{or } l + 2m + 0n = 0 \quad \text{(iii)}$$

Solving (i) and (ii), we get

$$\frac{l}{1} = \frac{m}{-2} = \frac{n}{-2}$$

So the direction cosines of the first line are proportional to 1, -2, -2.

Similarly, solving (i) and (iii), we get

$$\frac{l}{2} = \frac{m}{-1} = \frac{n}{2}$$

which means that the direction cosines of the second line are proportional to 2, -1, 2.

If θ be the angle between the lines, then

$$\cos \theta = \pm \frac{(2)(1) + (-1)(-2) + (2)(-2)}{\sqrt{4+1+4}\sqrt{1+4+4}} = 0 \Rightarrow \theta = \frac{\pi}{2}.$$

Thus the given straight lines are at right angles.

7. If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the direction cosines of two mutually perpendicular lines, show that the direction cosines of the line perpendicular to them are

$$m_1 n_2 - m_2 n_1, n_1 l_2 - n_2 l_1, l_1 m_2 - l_2 m_1.$$

Solution: Let (l, m, n) be the direction cosines of the required line. If it is perpendicular to two mutually perpendicular lines with direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) , we have $ll_1 + mm_1 + nn_1 = 0$ and $ll_2 + mm_2 + nn_2 = 0$. We can write

$$\frac{l}{m_1 n_2 - m_2 n_1} = \frac{m}{n_1 l_2 - n_2 l_1} = \frac{n}{l_1 m_2 - l_2 m_1}.$$

Hence the actual direction cosines of the required line are

$$\frac{m_1 n_2 - m_2 n_1}{\sqrt{(m_1 n_2 - m_2 n_1)^2}}, \frac{n_1 l_2 - n_2 l_1}{\sqrt{(n_1 l_2 - n_2 l_1)^2}}, \frac{l_1 m_2 - l_2 m_1}{\sqrt{(l_1 m_2 - l_2 m_1)^2}}.$$

Now we also have $\sin \theta = \sqrt{(m_1 n_2 - m_2 n_1)^2}$, where θ is the angle between the lines with direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) . Since the lines are mutually perpendicular, we have

$$\sqrt{(m_1 n_2 - m_2 n_1)^2} = 1.$$

Hence, $(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2 = 1$, so that the direction cosines are $m_1n_2 - m_2n_1, n_1l_2 - n_2l_1, l_1m_2 - l_2m_1$.

8. Find the direction cosines of the line drawn from the origin to the point $(-6, 2, 3)$.

Solution: Let l, m, n be the dcs of the line drawn from the origin to $(-6, 2, 3)$.

$$\text{So that } l = \frac{-6 - 0}{\sqrt{(-6 - 0)^2 + (2 - 0)^2 + (3 - 0)^2}} = \frac{-6}{\sqrt{49}} = \frac{-6}{7},$$

$$m = \frac{2 - 0}{\sqrt{36 + 4 + 9}} = \frac{2}{\sqrt{49}} = \frac{2}{7} \text{ and } n = \frac{3 - 0}{\sqrt{36 + 4 + 9}} = \frac{3}{\sqrt{49}} = \frac{3}{7}$$

Thus the required dcs are $\frac{-6}{7}, \frac{2}{7}$ and $\frac{3}{7}$.

9. Find the direction cosines of the line which is equally inclined to the axes.

Solution: Suppose that the line makes angles α, β, γ with the respective coordinate axes. Given that $\alpha = \beta = \gamma$. So that $\cos \alpha = \cos \beta = \cos \gamma$, i.e., $l = m = n$.

$$\text{Now } \frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{1+1+1}} = \pm \frac{1}{\sqrt{3}}.$$

Thus the required dcs are $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$.

10. Prove that the two lines whose direction cosines are connected by the relations

$$al + bm + cn = 0 \text{ and } ul^2 + vm^2 + wn^2 = 0 \text{ are perpendicular if}$$

$$a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$$

$$\text{and are parallel if } \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0.$$

Solution: We have $al + bm + cn = 0$ (1)

$$\text{and } ul^2 + vm^2 + wn^2 = 0 \quad (2)$$

Eliminating n from (1) and (2), we get

$$ul^2 + vm^2 + w\left(\frac{al + bm}{-c}\right)^2 = 0$$

$$\Rightarrow (c^2u + a^2w)\frac{l^2}{m^2} - 2abw\frac{l}{m} + (b^2w + c^2v) = 0 \quad (3)$$

Let $\frac{l_1}{m_1}$ and $\frac{l_2}{m_2}$ be the roots of (3). Then $\frac{l_1}{m_1} \cdot \frac{l_2}{m_2} = \frac{b^2w + c^2v}{c^2u + a^2w} \Rightarrow \frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w}$

Similarly, eliminating l from (1) and (2) we get $\frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{a^2v + b^2u}$

Thus, we obtain the relation $\frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{a^2v + b^2u}$.

(l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of the given lines. If the lines are perpendicular then $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$.

$$\Rightarrow (b^2 w + c^2 v) + (c^2 u + a^2 w) + (a^2 v + b^2 u) = 0$$

$$\Rightarrow a^2(v+w) + b^2(w+u) + c^2(u+v) = 0.$$

If the lines are parallel then their corresponding direction cosines are equal and consequently, the

roots of (3) are equal, i.e., $(-2abw)^2 - 4(c^2 u + a^2 w)(b^2 w + c^2 v) = 0$

$$\Rightarrow 4a^2 b^2 w^2 - 4b^2 c^2 uw - 4c^4 uv - 4a^2 b^2 w^2 - 4a^2 c^2 vw = 0$$

$$\Rightarrow b^2 c^2 uw + c^4 uv + a^2 c^2 vw = 0$$

$$\Rightarrow \frac{b^2}{v} + \frac{c^2}{w} + \frac{a^2}{u} = 0 \Rightarrow \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0.$$

Exercises

1. Find the angle between the lines whose direction cosines are given by the equations

$$l + m + n = 0 \text{ and } l^2 + m^2 - n^2 = 0.$$

2. Show that the lines whose direction cosines are given by the equations $l + m + n = 0$ and $al^2 + bm^2 + cn^2 = 0$ are perpendicular if $a + b + c = 0$ and parallel if

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0.$$

3. Prove that the two lines whose direction cosines are given by the relations

$$al + bm + cn = 0 \text{ and } fmn + gnl + hlm = 0$$

are perpendicular if $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ and are parallel if

$$\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0.$$

Reference Books

1. An Elementary Treatise on Coordinate Geometry of Three Dimensions by Robert J. T. Bell
2. The Elements of Coordinate Geometry by S. L. Loney
3. Solid Geometry by M. L. Khanna